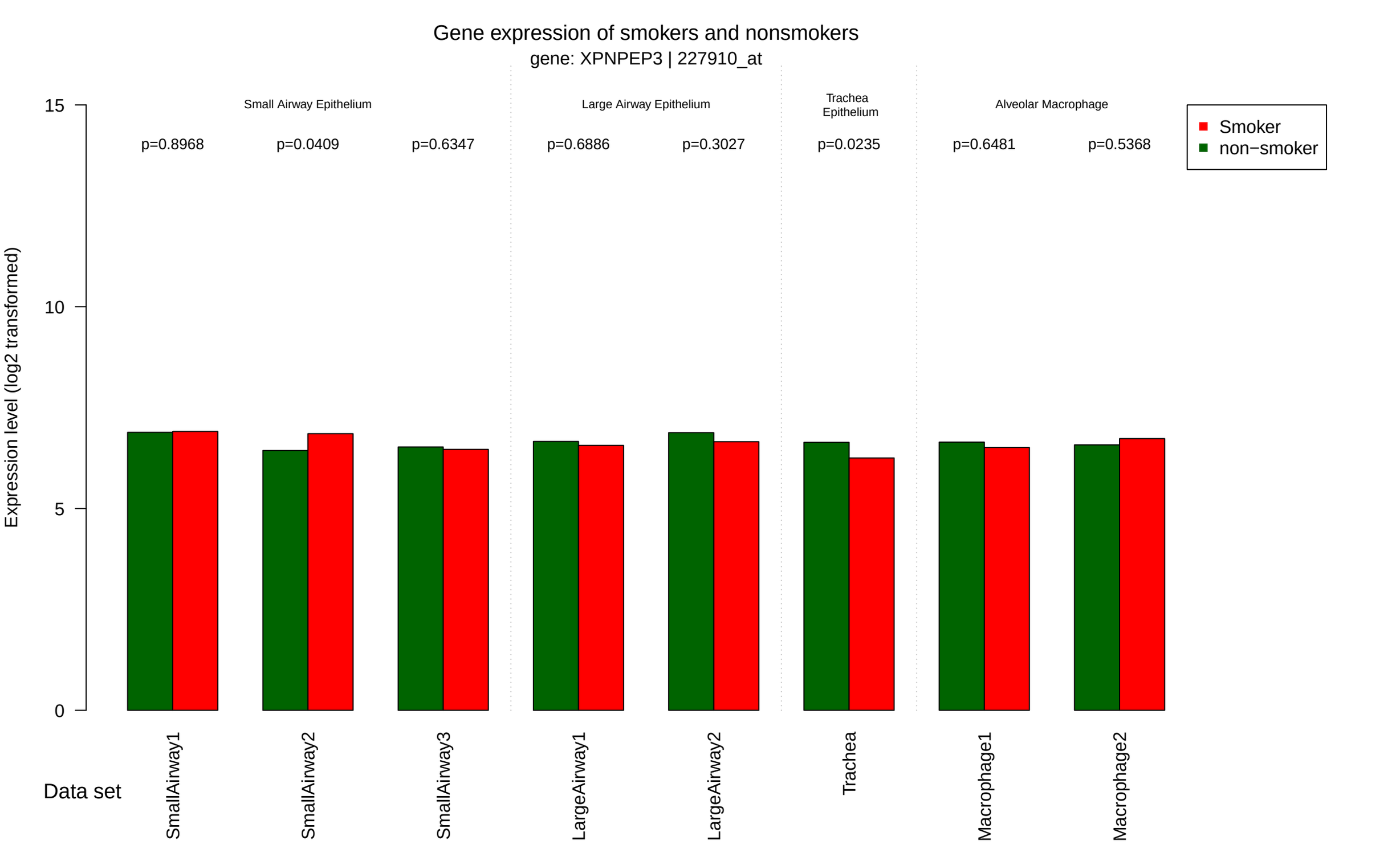
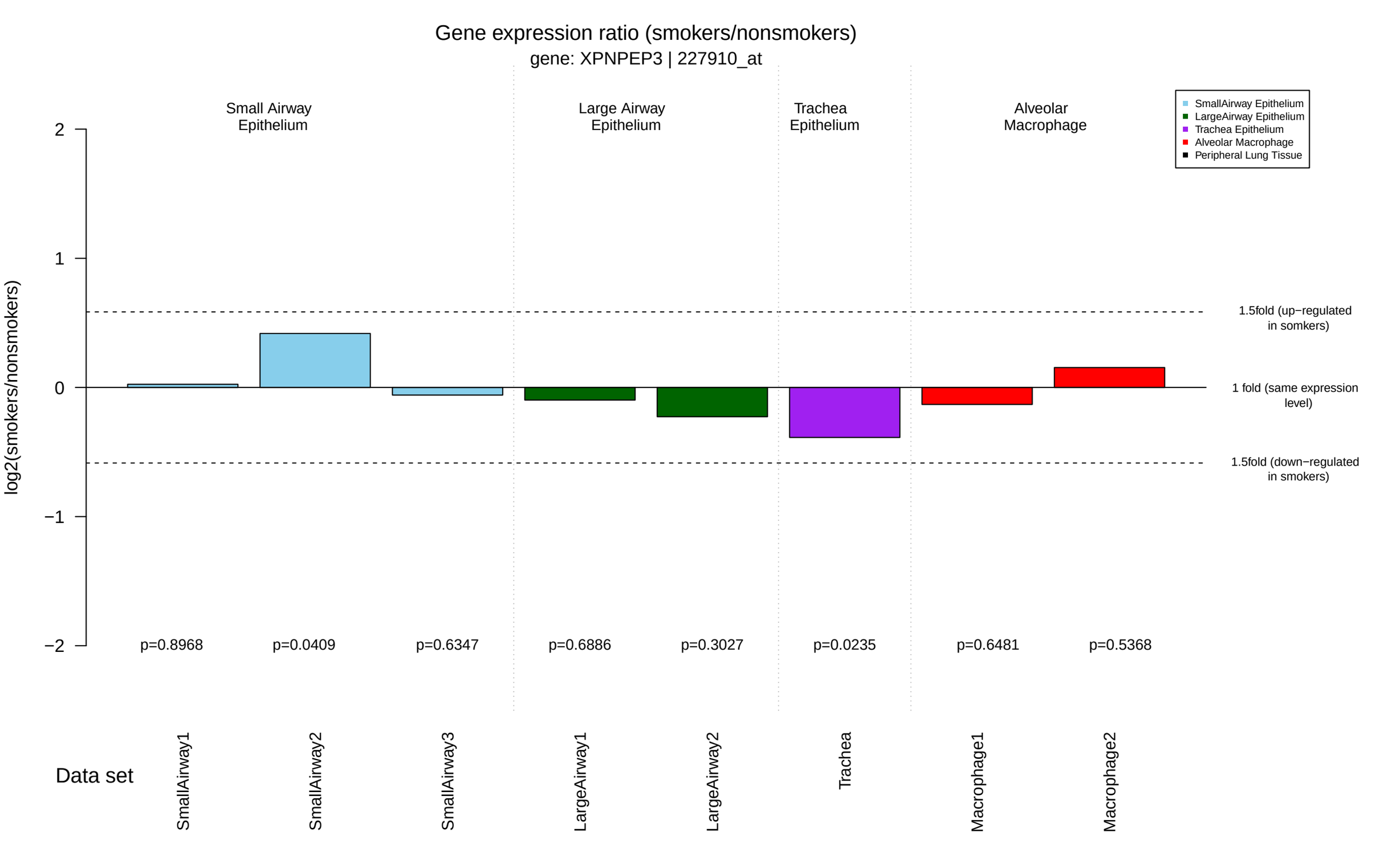
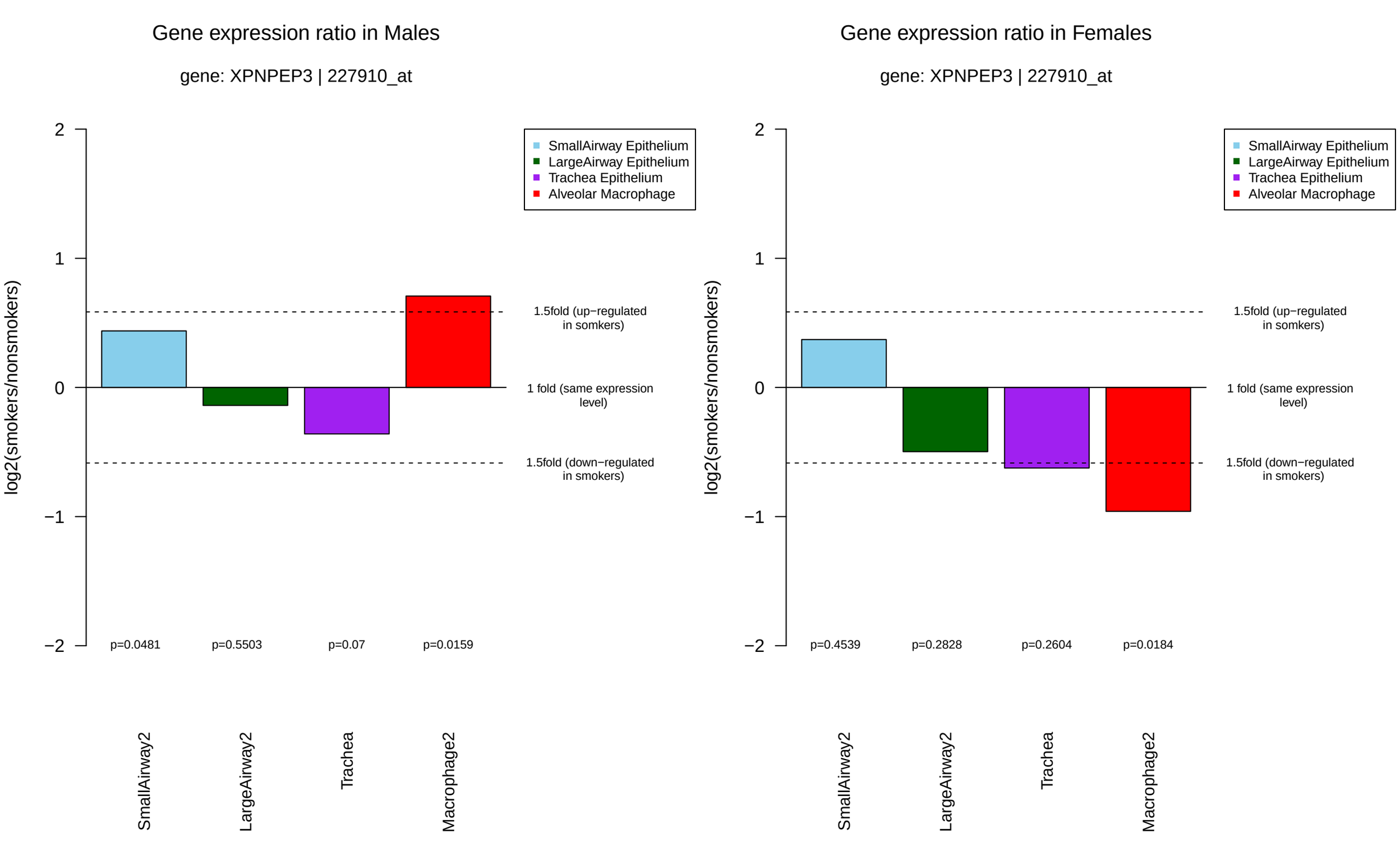
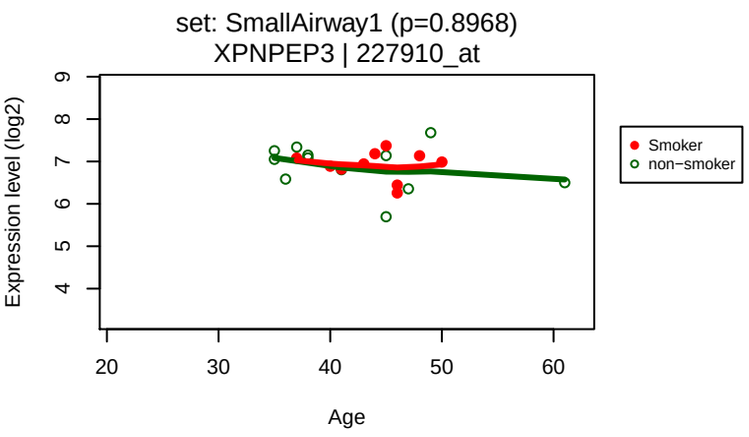
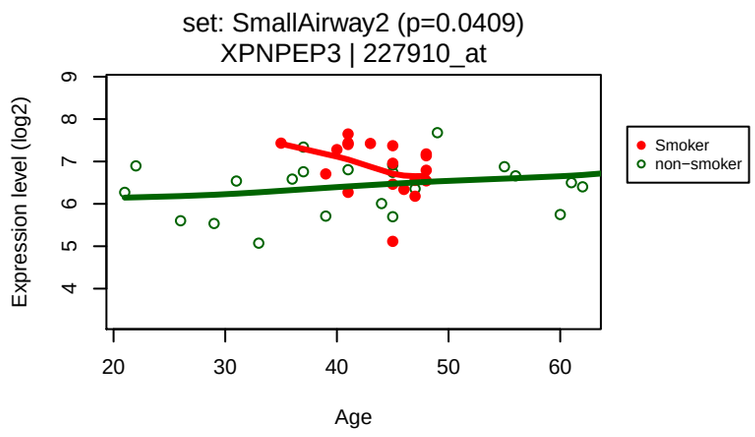
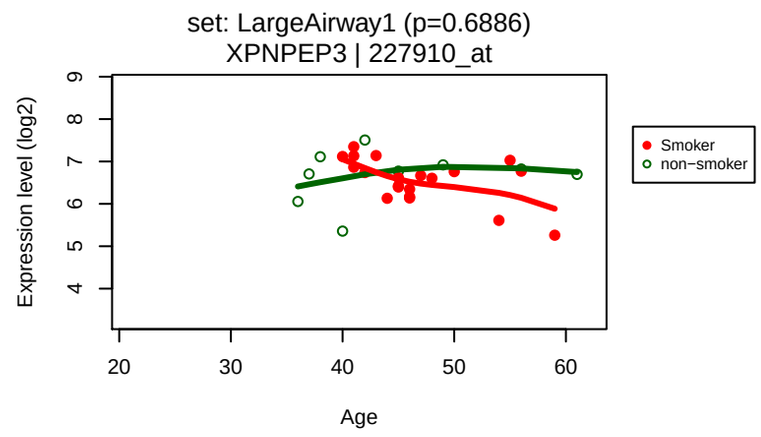

Gene expression of smokers and nonsmokers
gene: XPNPEP3 | 227910_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8968
p=0.0409
p=0.6347
p=0.6886
p=0.3027
p=0.0235
p=0.6481
p=0.5368
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

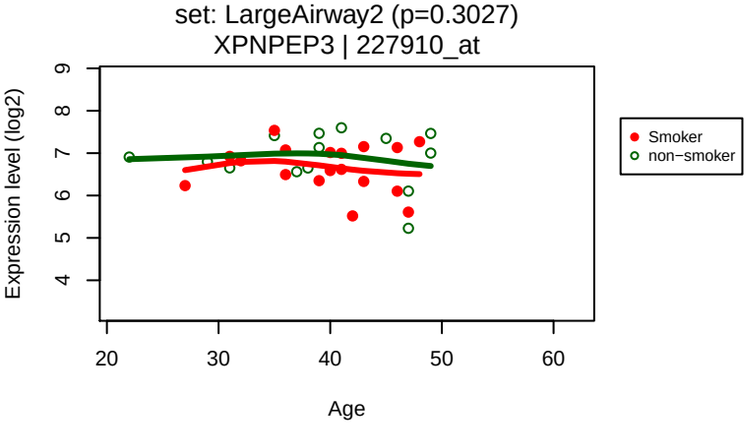

Gene expression ratio (smokers/nonsmokers)
gene: XPNPEP3 | 227910_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8968
p=0.0409
p=0.6347
p=0.6886
p=0.3027
p=0.0235
p=0.6481
p=0.5368
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

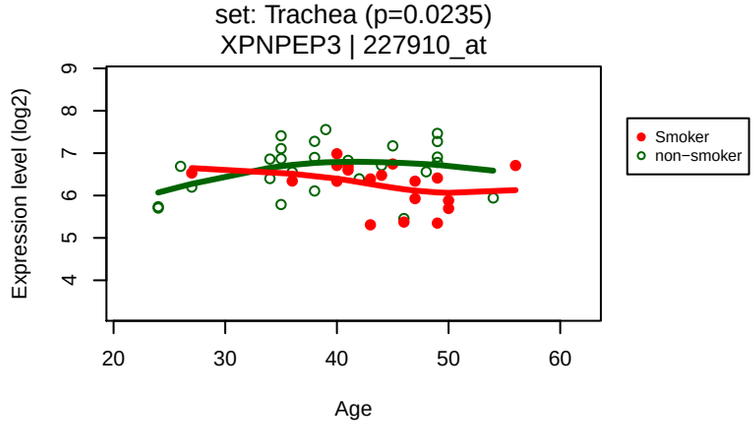

Gene expression ratio in Males
gene: XPNPEP3 | 227910_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0481
p=0.5503
p=0.07
p=0.0159
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: XPNPEP3 | 227910_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4539
p=0.2828
p=0.2604
p=0.0184
SmallAirway2
LargeAirway2
Trachea
Macrophage2

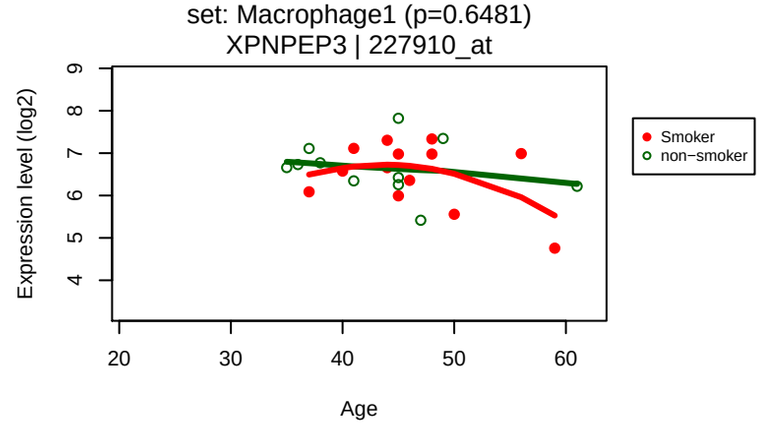

set: SmallAirway1 (p=0.8968)
XPNPEP3 | 227910_at
Expression level (log2)
Age
Smoker
non−smoker

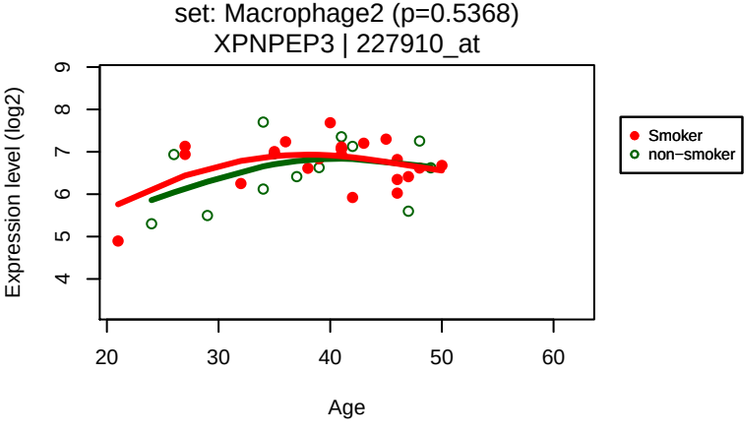

set: SmallAirway2 (p=0.0409)
XPNPEP3 | 227910_at
Expression level (log2)
Age
Smoker
non−smoker

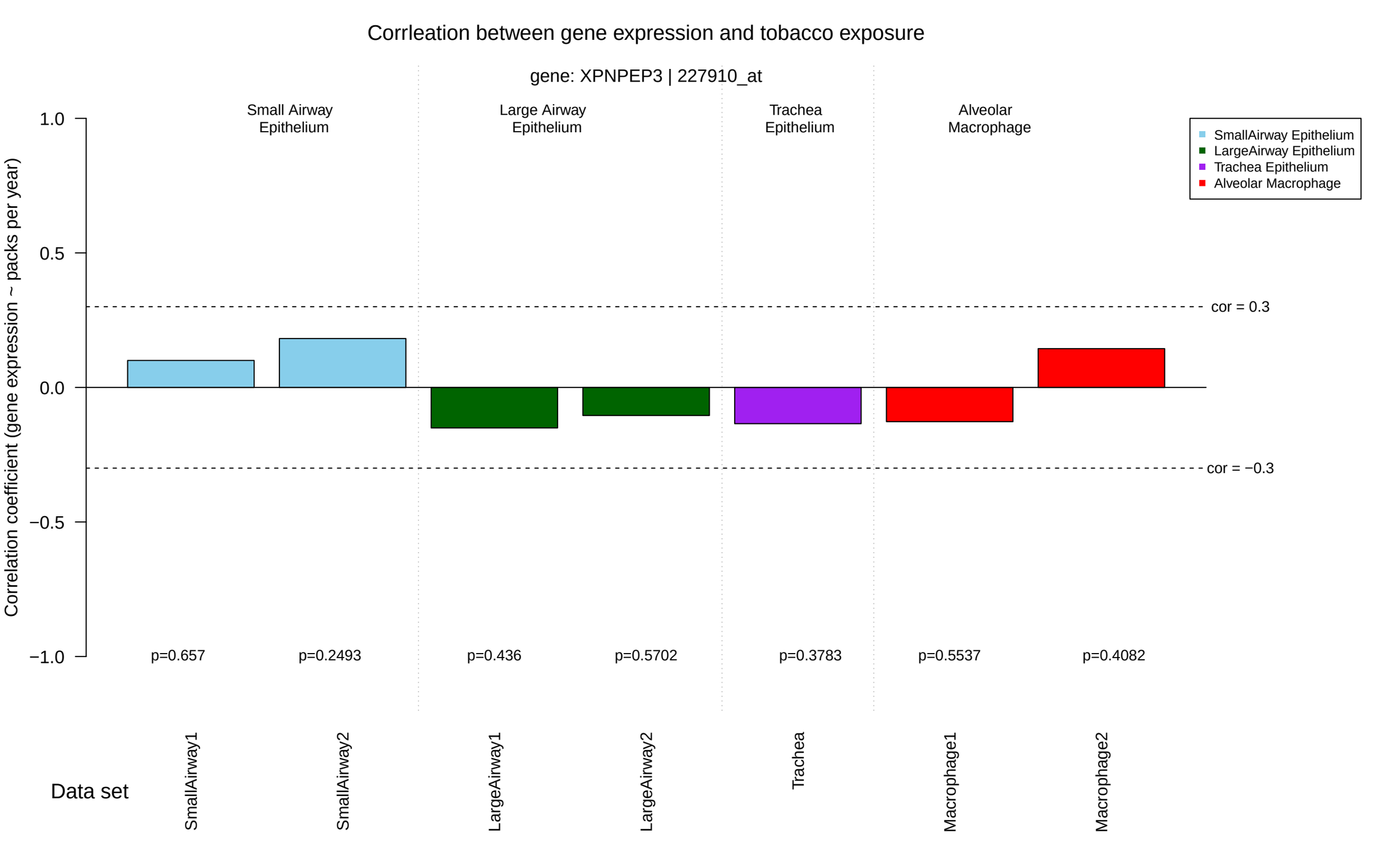

set: LargeAirway1 (p=0.6886)
XPNPEP3 | 227910_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3027)
XPNPEP3 | 227910_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0235)
XPNPEP3 | 227910_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6481)
XPNPEP3 | 227910_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5368)
XPNPEP3 | 227910_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: XPNPEP3 | 227910_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.657
p=0.2493
p=0.436
p=0.5702
p=0.3783
p=0.5537
p=0.4082
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2